

An Innovative Strategy for the Synthesis of Single-Atom Catalysts on MXene Surfaces

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Our Goal

- ❑ We suggest a method to **enhance the stability of single-atom catalysts on Ti₃C₂T_x MXene** by **esterifying** its surface with hemin, followed by melamine addition and pyrolysis.
- ❑ This process fixes Fe atoms as single atoms, **improving catalyst durability and oxidation resistance** during water electrolysis.

Introduction

Ti₃C₂T_x MXene

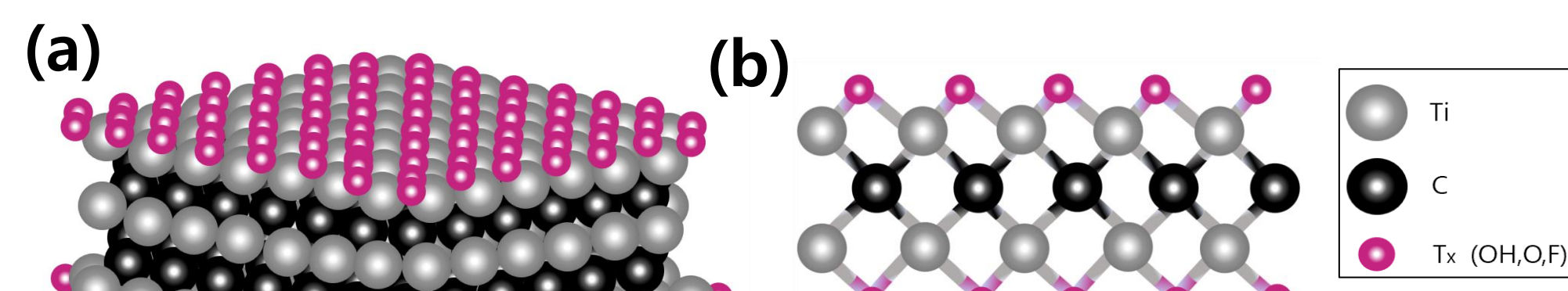


Figure 1. (a) The scheme of Ti₃C₂T_x MXene
(b) Hydrophilic property of MXene surface(62.47±2.77°)

■ Various applications

- ▶ Electrochemical catalysis
- ▶ Water electrolysis

■ 2D transition metal carbide

- ▶ Electrical conductivity ↑ & hydrophilicity ↑

■ Synthesis of Ti₃C₂T_x MXene

- ▶ Based on anhydrous etching and delamination

Single-Atom Catalysts

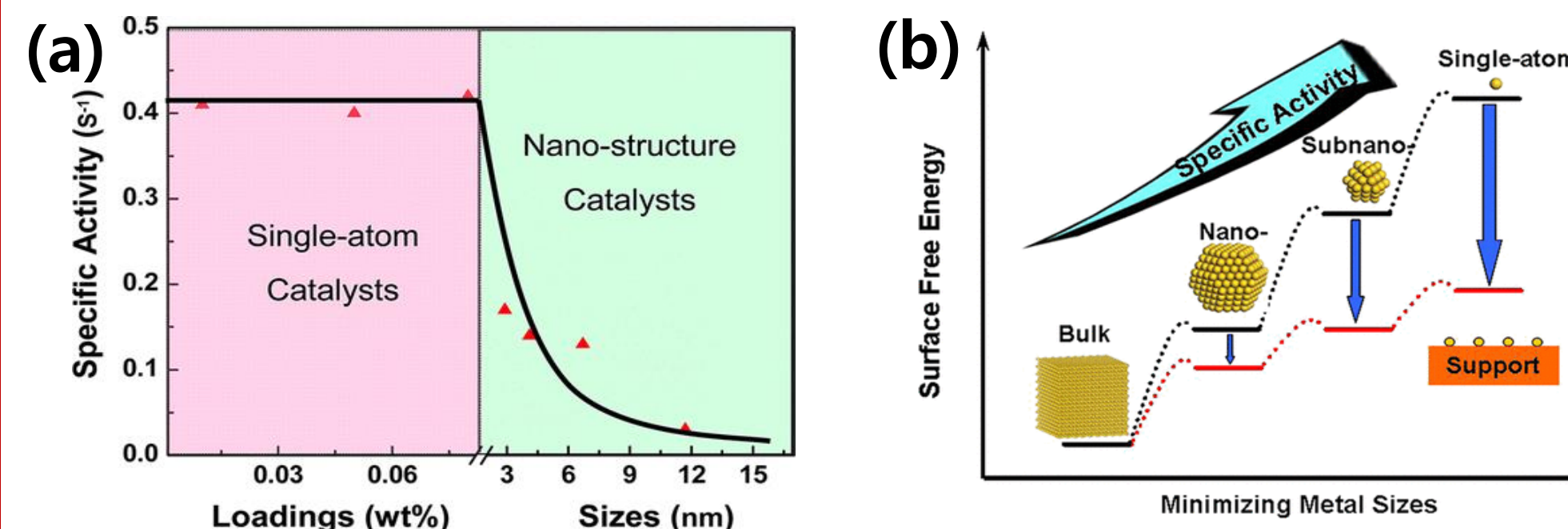


Figure 2. (a) A graph showing the superior specific activity of SACs

■ Excellent Specific Activity(s⁻¹)

- ▶ ↑ than nano-structure catalysts
- ▶ Higher ORR performance

■ Importance of substrate of SAC

- ▶ Affects catalyst's lifetime and stability of isolated metal atoms
- ▶ Ti₃C₂T_x MXene can be a support material for SACs

Mxene substrate for Single Atom Catalysts

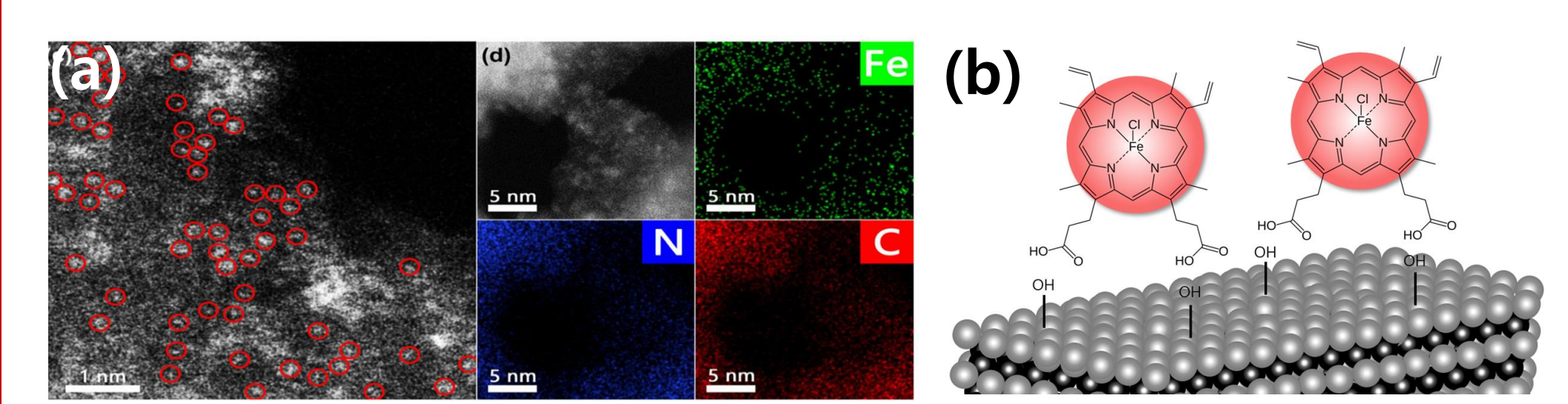


Figure 3 [1]. (a) EDS mapping of Acid-treated-BP2000 grafting hemin by esterification. (b) The scheme of hemin and MXene for esterification

■ Esterification

- ▶ Carboxylic acid (hemin) + Alcohol (MXene surface)
- ▶ anchored hemin on MXene

■ Pyrolysis of esters

- ▶ anchored single Fe atoms on MXene

Current Issues

■ Conventional methods of synthesizing SACs on Ti₃C₂T_x

- ✓ Instability in nano-chemical synthesis
- ✓ Aggregation of metal atoms & oxidation

Our Approach

■ Esterification with hemin

- ✓ Addition of melamine as spacer: Prevent aggregation → subsequent pyrolysis
- ✓ Facilitates the fixation of Fe atoms as single atoms onto MXene
- ✓ Condensation reaction of melamine into g-C₃N₄.

Experimental Section

Synthesis of MXene

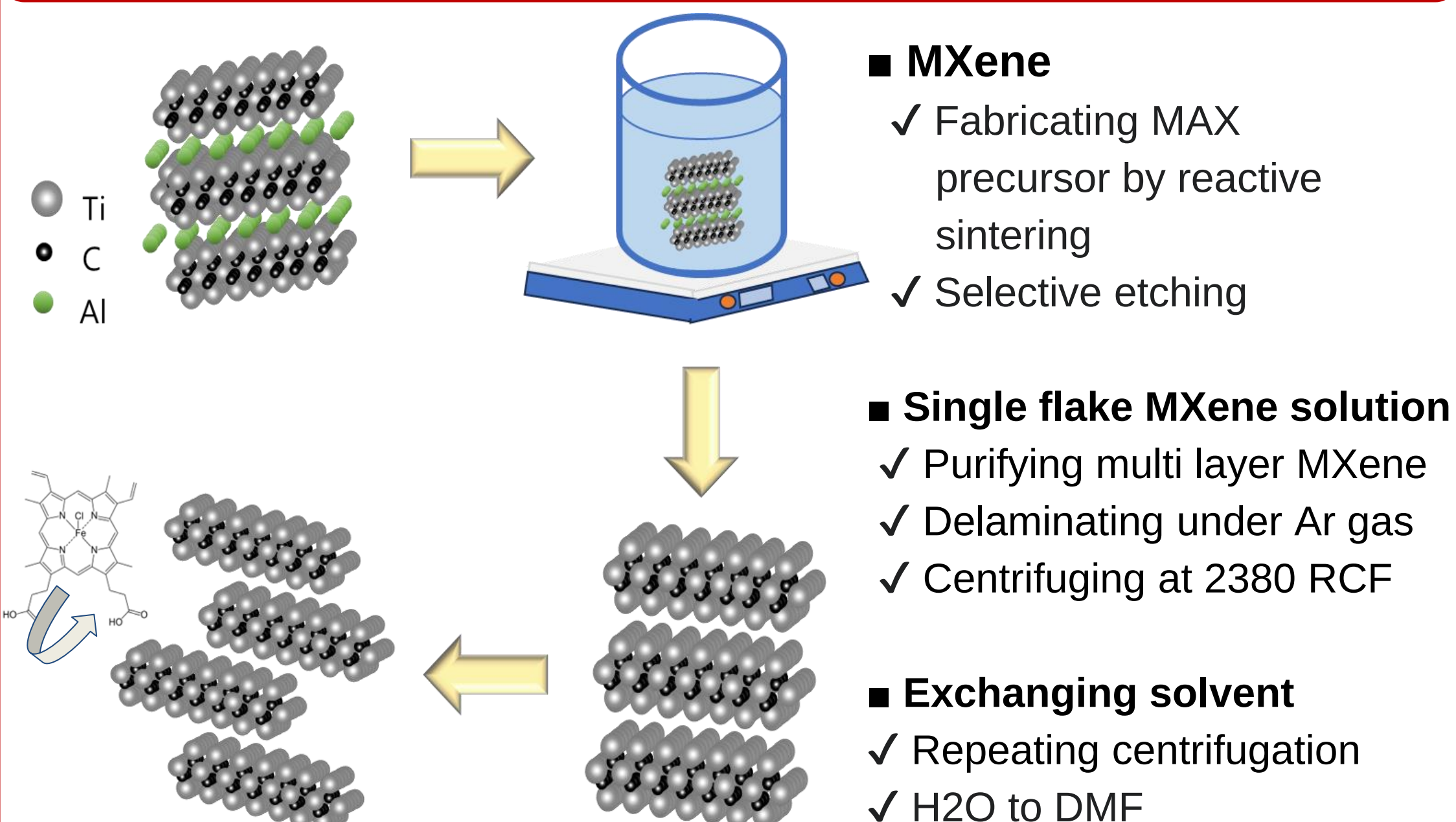


Figure 4. The scheme of Ti₃C₂T_x MXene synthesis

Esterification reaction

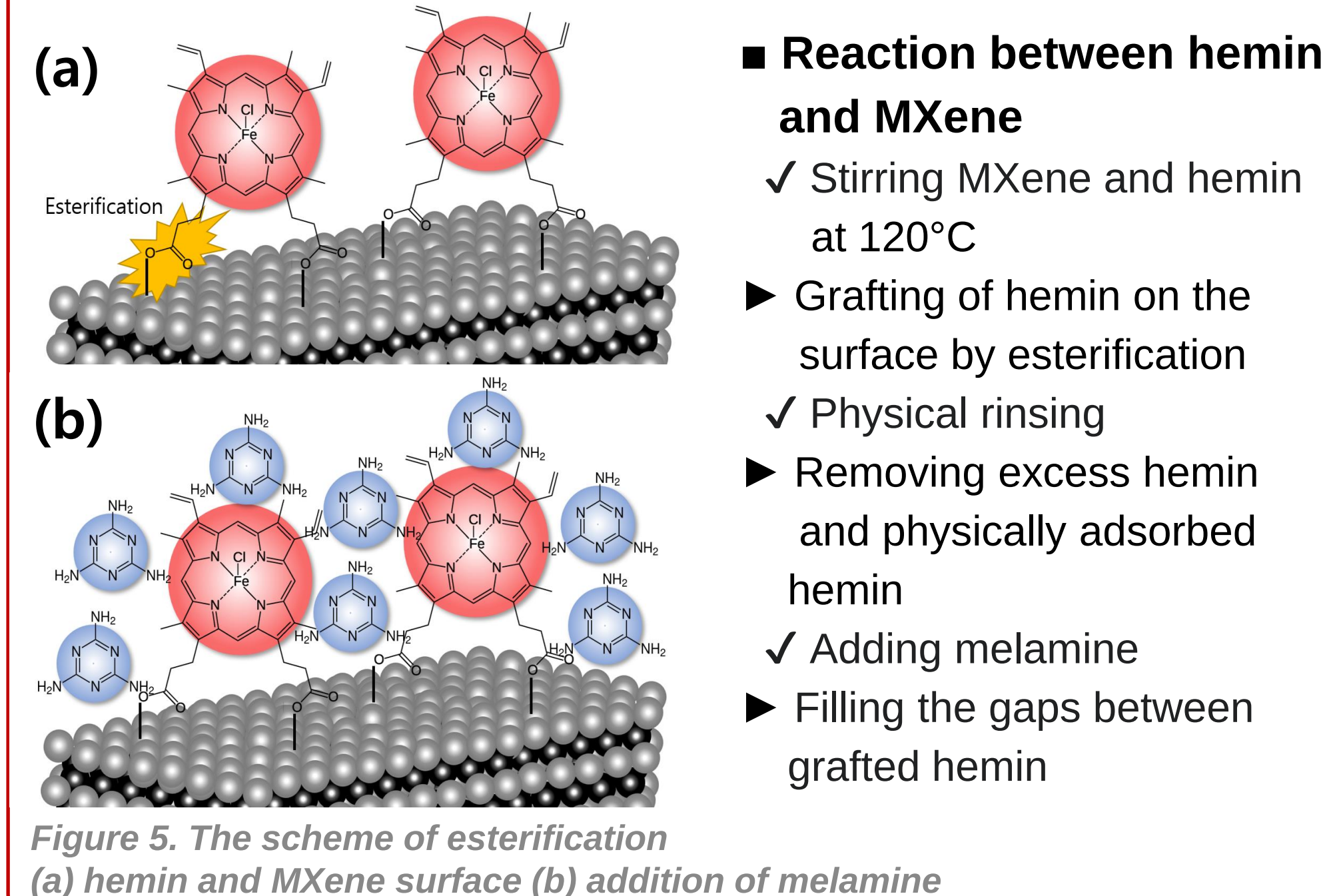


Figure 5. The scheme of esterification
(a) hemin and MXene surface (b) addition of melamine

Pyrolysis of hemin

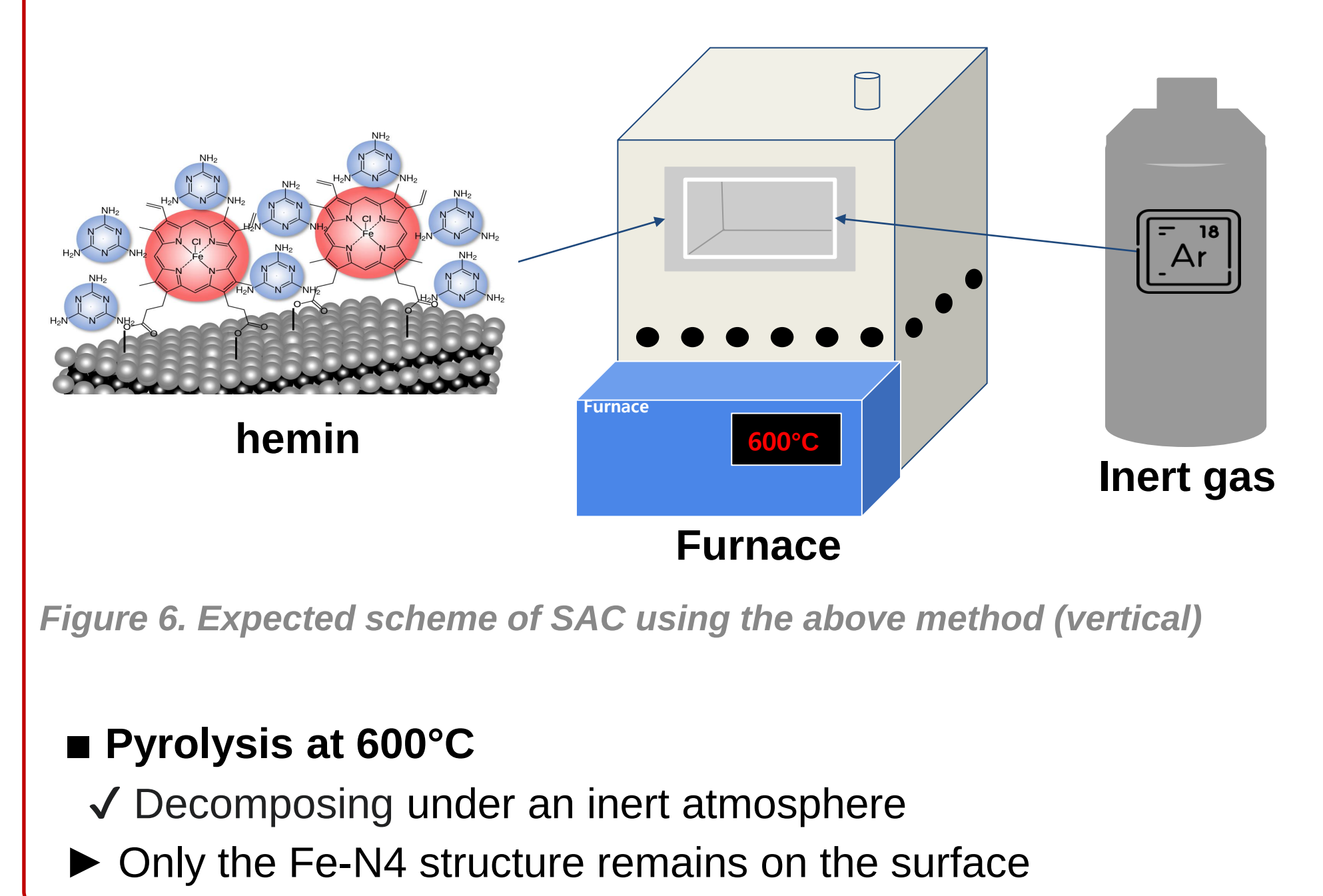


Figure 6. Expected scheme of SAC using the above method (vertical)

Result & Discussion

Expected figure of FeN₄

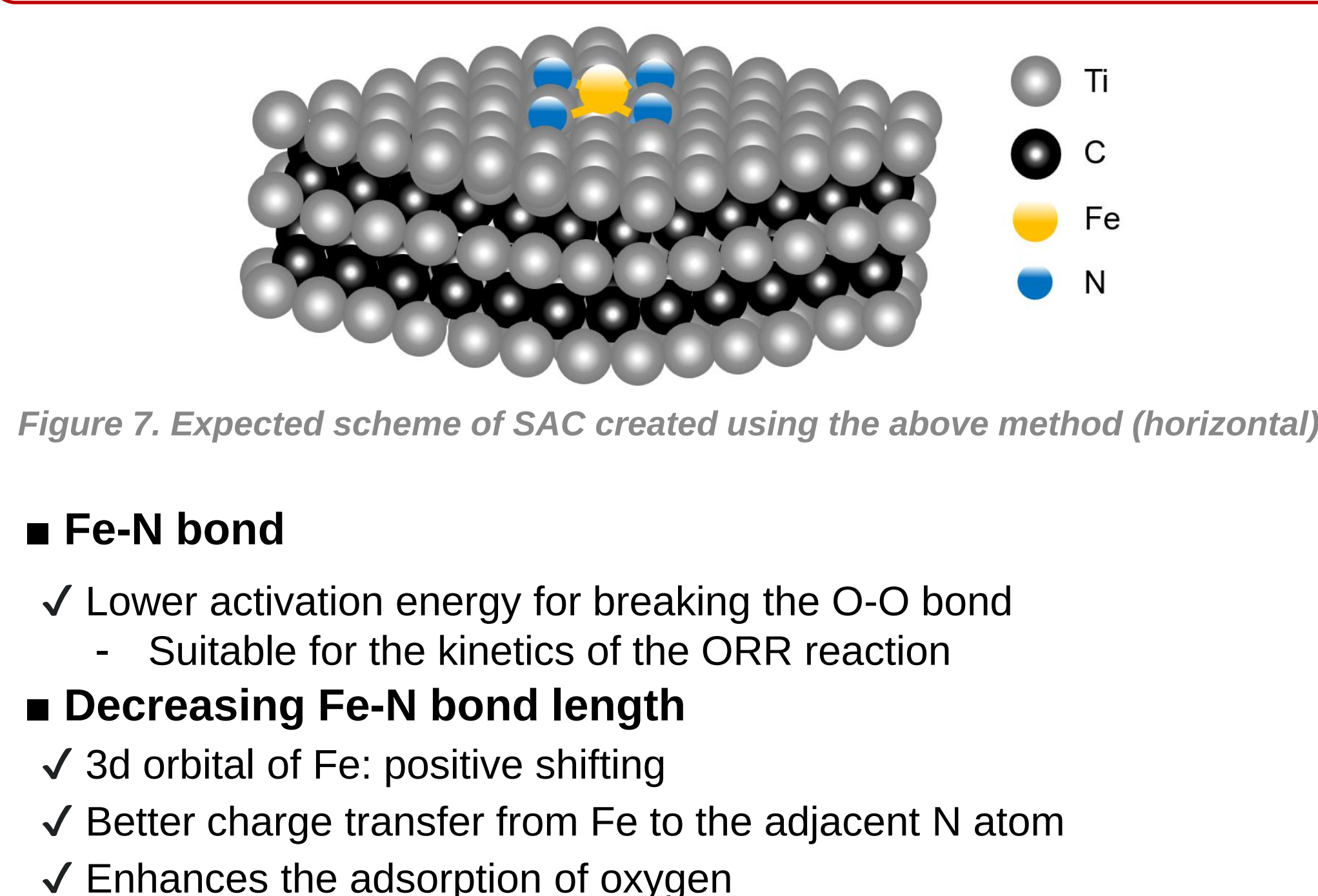


Figure 7. Expected scheme of SAC created using the above method (horizontal)

Superior ORR properties

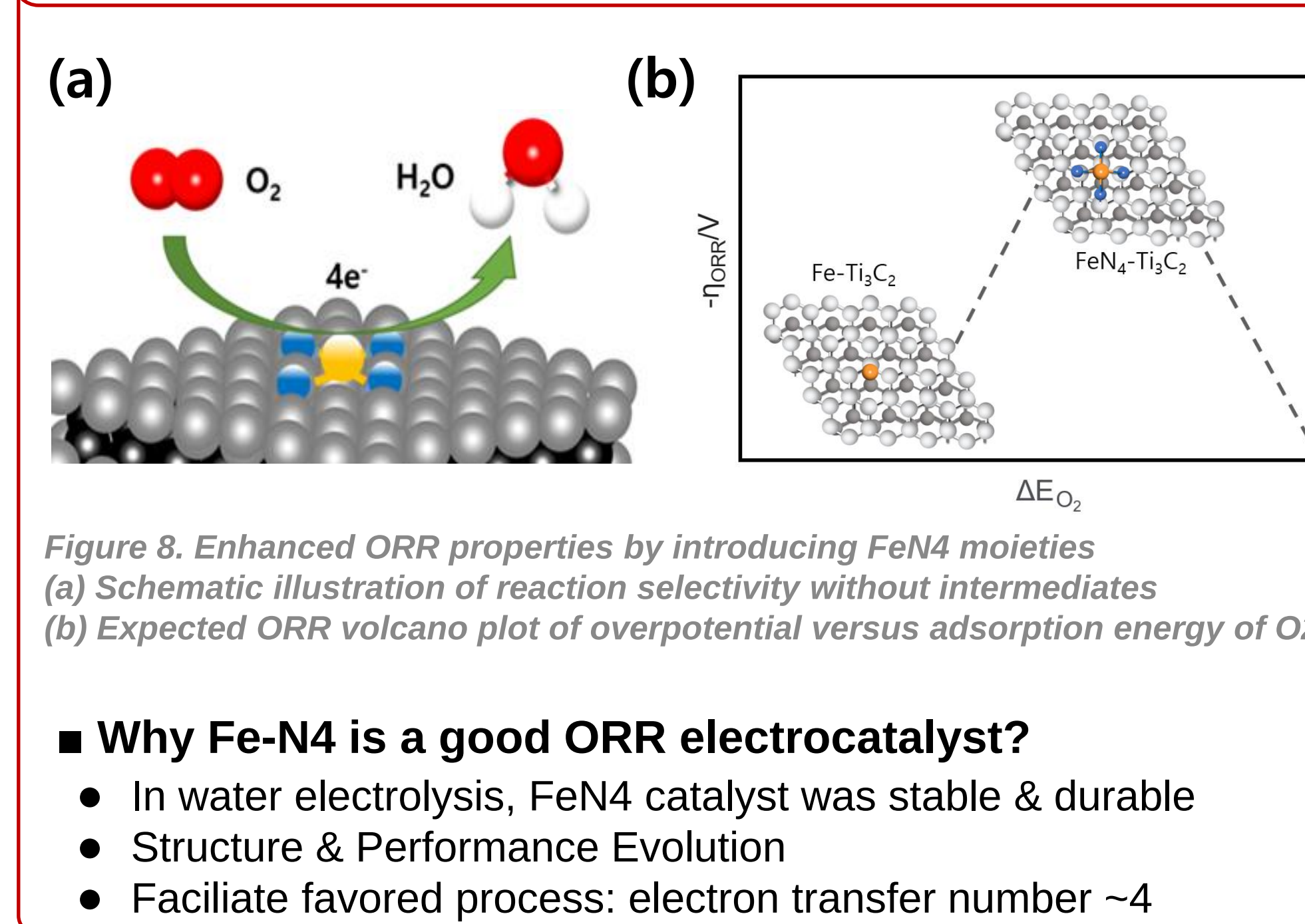


Figure 8. Enhanced ORR properties by introducing FeN₄ moieties
(a) Schematic illustration of reaction selectivity without intermediates
(b) Expected ORR volcano plot of overpotential versus adsorption energy of O₂

Pyrolysis of hemin

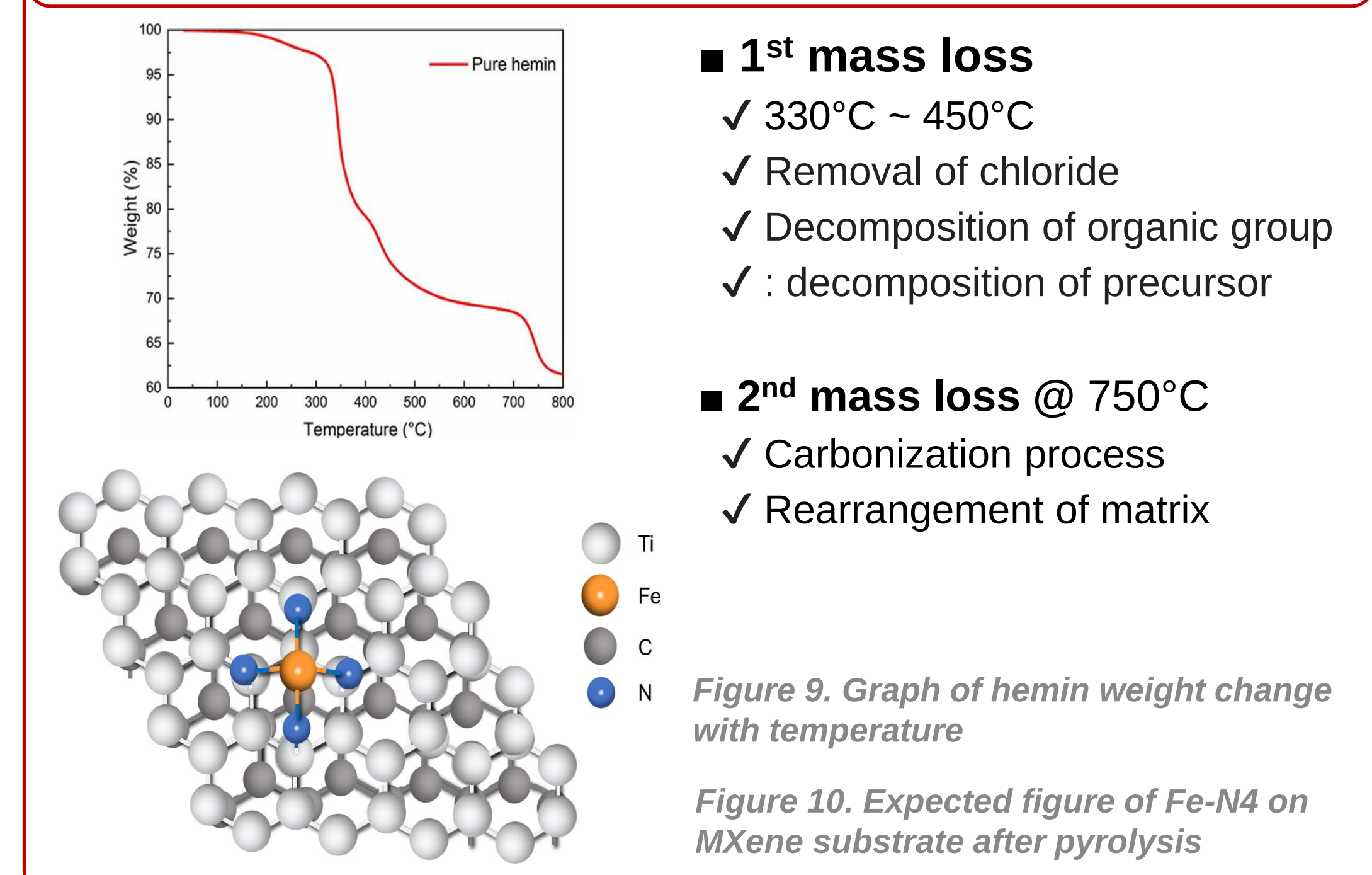


Figure 9. Graph of hemin weight change with temperature

Figure 10. Expected figure of Fe-N₄ on MXene substrate after pyrolysis

Conclusion

- By creating N₄ sites using pyrolysis of meso-Tetra (4-carboxyphenyl)porine (hemin), it is expected that locally stable terrace sites will form, which could enable more stable behavior when the catalyst is activated for electrocatalytic reactions
- Melamine will serve as a spacer, an N atom dopant. During the pyrolysis process, it will condense into g-C₃N₄, anchoring the iron atoms from the decomposing hemin. As melamine decomposes, a reducing atmosphere will be created and it can prevent the oxidation of MXene
- This method will not only enhances the stability of single-atom catalysts on the MXene surface but also protects them against oxidation, ensuring the catalysts remain active and durable during water electrolysis
- This surface modification strategy effectively prevents metal atom aggregation and oxidation
- The resulting Fe single-atom catalysts on MXene show promise for use in water electrolysis, potentially leading to more efficient and durable catalytic systems

Future Plan

- We will continue experiments based on the study to verify the feasibility and effectiveness of our study, including pyrolysis of MXene higher than 500°C

■ References

- [1] Delgado, S., Remedios-Díaz, Y., Castilla-Silvestre, A. C., González-Rodríguez, M., García, G., & Pastor, E. (2024). MXenes in electrocatalysis. In MXenes as Surface-Active Advanced Materials (pp. 317-336). Elsevier.
- [2] Ding, R., Liu, Y., Rui, Z., Li, J., Liu, J., & Zou, Z. (2020). Facile grafting strategy synthesis of single-atom electrocatalyst with enhanced ORR performance. Nano Research, 13, 1519-1526.